HSDSD Exam Prep: Key Formulas and Worked Examples

1 Exam Prep: Key Formulas and Worked Examples

1.1 Part 1 Key Formula Sheet

1.1.1 Chapter 1

Formula	Meaning	Reference
$f_{\max} = 0.32/\tau_r$	Highest frequency of interest	chap1.pdf p.6
$T_D \geq t_r/10$	TX line criterion	chap1.pdf p.7

1.1.2 Chapter 2

Formula	Meaning	Reference
$Z_0 = \sqrt{L/C}$	Characteristic impedance (lossless)	chap2.pdf p.9
$Z_0 \approx \frac{87}{\sqrt{\varepsilon_r+1.41}} \ln \frac{5.98H}{0.8W+T}$	Microstrip impedance	chap2.pdf p.9
$Z_0 \approx \frac{60}{\sqrt{\varepsilon_r}} \ln \frac{4H}{0.67\pi(T+0.8W)}$	Symmetric stripline impedance	chap2.pdf p.10
$v = c/\sqrt{\varepsilon_r}$, $\mathrm{TD} = x\sqrt{\varepsilon_r}/c$	Propagation velocity and delay	chap2.pdf p.11
$\mathrm{TD} = \sqrt{LC}$, $Z_0 = \sqrt{L/C}$	Alternate forms $L = \mathrm{TD} \cdot Z_0$, $C = \mathrm{TD}/Z_0$	chap2.pdf p.11
$\varepsilon_e = \frac{\varepsilon_r+1}{2} + \frac{\varepsilon_r-1}{2}(1+\frac{12H}{W})^{-1/2} + F - 0.217(\varepsilon_r-1)\frac{T}{\sqrt{WH}}$	Effective ε for microstrip	chap2.pdf p.12
Segments $\geq 10 \cdot \mathrm{TD}/T_r$	SPICE model segmentation	chap2.pdf p.14
$V_i = V_s Z_0/(Z_0 + Z_s)$	Initial wave voltage	chap2.pdf p.17
$\rho = (Z_t - Z_0)/(Z_t + Z_0)$	Reflection coefficient	chap2.pdf p.18

1.1.3 Chapter 3

Formula	Meaning	Reference
$V_{\text{noise}} = L_m\, dI/dt$	Inductive coupling noise	chap3.pdf p.4
$I_{\text{noise}} = C_m\, dV/dt$	Capacitive coupling noise	chap3.pdf p.5
$A_{\text{NEXT}} = \frac{V_{\text{in}}}{4}(\frac{L_m}{L} + \frac{C_m}{C})$	NEXT peak (matched, $T_r < 2\mathrm{TD}$)	chap3.pdf p.13
$B_{\text{FEXT}} = -\frac{V_{\text{in}} X\sqrt{LC}}{2T_r}(\frac{L_m}{L} - \frac{C_m}{C})$	FEXT peak (matched, $T_r < 2\mathrm{TD}$)	chap3.pdf p.13
$Z_{\text{odd}} = \sqrt{(L_{11} - L_{12})/(C_{11} + C_m)}$	Odd-mode impedance	chap3.pdf p.27
$Z_{\text{even}} = \sqrt{(L_{11} + L_{12})/(C_{11} - C_m)}$	Even-mode impedance	chap3.pdf p.29
$K = L_{12}/\sqrt{L_{11}L_{22}}$	Inductive coupling factor	chap3.pdf p.21

1.1.4 Chapter 5

Formula	Meaning	Reference
$\delta = \sqrt{\rho/(\pi F \mu)}$	Skin depth	chap5.pdf p.6
$R_{\text{ac}} \approx \sqrt{\rho\pi\mu F}(1/W + 1/6H)$	AC resistance, microstrip (approx)	chap5.pdf p.12

Formula	Meaning	Reference
$R_{\text{ac}} = R_S\sqrt{F}$	AC resistance via surface resistance	chap5.pdf p.15
$\tan\delta_d = \varepsilon''/\varepsilon'$	Dielectric loss tangent	chap5.pdf p.21
$G = \tan\delta_d \cdot 2\pi F C_{11}$	Dielectric shunt conductance	chap5.pdf p.21
$C_{90} \approx C_{11} \cdot w$	Excess capacitance at 90ř bend	chap5.pdf p.39

1.2 Part 2 Worked Example: Chapter 2

1.2.1 Problem (from chap2.pdf, pp.3743)

A driver ($Z_s = 30\,\Omega$, edge rate $t_r = 100$ ps, swing 0 to 2 V) drives a microstrip trace ($Z_0 = 50\,\Omega$, length 5 in., $\varepsilon_r = 4.0$, $T = 1.0$ mil, $W = 5$ mil).

Step 1: Find H (trace height)

Using $Z_0 = 50\,\Omega$ in the microstrip formula:

$$50 = \frac{87}{\sqrt{4.0+1.41}} \ln\frac{5.98H}{0.8(5)+1.0} \implies H = 3.2 \text{ mils}$$

(chap2.pdf, p.39)

Step 2: Find effective ε_e

$W/H = 5/3.2 = 1.5625 > 1$, so $F = 0$.

$$\varepsilon_e = \frac{4+1}{2} + \frac{4-1}{2}\left(1+\frac{12(3.2)}{5}\right)^{-1/2} + 0 - 0.217(4-1)\frac{1.0}{\sqrt{5.0\times 3.2}} = 2.84$$

(chap2.pdf, p.40)

Step 3: Find propagation velocity and TD

$$v = \frac{3\times 10^8}{\sqrt{2.84}} = 1.78\times 10^8 \text{ m/s}$$

$$\text{TD} = \frac{5 \text{ in.} \times 0.0254 \text{ m/in.}}{1.78\times 10^8 \text{ m/s}} = 713 \text{ ps}$$

(chap2.pdf, p.40)

Step 4: Determine wave shape (lattice diagram)

Initial voltage: $V_i = 2\times 50/(50+30) = 1.25$ V

$\rho_{\text{load}} = (-50)/(+50) = +1$ (open load assumed)

At $t = \text{TD}$: load sees $1.25 + 1.25(+1) = 2.5$ V

$\rho_{\text{source}} = (30-50)/(30+50) = -0.25$

At $t = 2\text{TD}$: source reflection $= 1.25\times(-0.25) = -0.3125$ V, so source sees $1.25 - 0.3125 = 0.9375$ V above steady state.

Steady state $= 2\times\infty/(\infty+30) \approx 2$ V (for open load). The ringing settles at 2 V.

(chap2.pdf, p.41)

Step 5: Number of SPICE segments

Since $t_r < 2\text{TD}$ (100 ps < 1426 ps), reflections matter:

$$N \geq 10\left(\frac{713 \text{ ps}}{100 \text{ ps}}\right) = 71.3 \implies \text{use } 72$$

(chap2.pdf, p.42)

1.3 Part 3 Worked Example: Chapter 3

1.3.1 Problem (from chap3.pdf, pp.1718 Example 3.2)

Two coupled lines: $Z_0 = 70\,\Omega$, both ends terminated at 70 , $V_{\text{in}} = 1.0$ V, $T_r = 100$ ps, length $X = 2$ in.

Given matrices:

$$[L] = \begin{bmatrix} L_{11} & L_m \\ L_m & L_{11} \end{bmatrix} \text{nH/in.}, \quad [C] = \begin{bmatrix} C_{11} & -C_m \\ -C_m & C_{11} \end{bmatrix} \text{pF/in.}$$

(Numerical values read from slide; extracted text was partially corrupted see chap3.pdf p.17 for exact matrix values.)

NEXT:

$$A_{\text{NEXT}} = \frac{1.0}{4}\left(\frac{L_m}{L_{11}} + \frac{C_m}{C_{11}}\right)$$

Check $T_r = 100$ ps vs. 2TD to determine which formula applies.

FEXT:

$$B_{\text{FEXT}} = -\frac{1.0 \times 2 \text{ in.}\sqrt{L_{11}C_{11}}}{2 \times 100 \text{ ps}}\left(\frac{L_m}{L_{11}} - \frac{C_m}{C_{11}}\right)$$

(chap3.pdf, p.18)

1.4 Part 4 Worked Example: Chapter 5

1.4.1 Problem (from chap5.pdf, pp.2327)

Stripline: $W = 5$ mil, $H_1 = H_2 = 10$ mil, $t = 0.63$ mil, $\varepsilon_r = 4.0$.

Step 1: Find R_S (surface resistance)

For copper: $\rho = 1.72 \times 10^{-8}$ ům, $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m.

$$R_S = \sqrt{\rho\pi\mu} = \sqrt{1.72 \times 10^{-8} \times \pi \times 4\pi \times 10^{-7}}$$

(chap5.pdf, p.24)

Step 2: Frequency where $\delta = t$

$$\delta = t = 0.63 \text{ mil} = 16\,\mu\text{m} \implies F_{\text{skin}} = \frac{\rho}{\pi\mu\delta^2}$$

Below this frequency: $R = R_{\text{DC}}$. Above: $R \propto \sqrt{F}$. This crossover is ˜17 MHz for copper with this geometry.

(chap5.pdf, p.26)

Step 3: AC resistance at 400 MHz

$$R_{\text{ac}} = R_S\sqrt{400 \times 10^6} \times \frac{1}{W} + \frac{1}{W_{\text{ground}}}$$

(chap5.pdf, p.27)

Step 4: Shunt conductance at 400 MHz

From PD $= \sqrt{\varepsilon_r}/c$:

$C_{11} = \text{PD}/Z_0 = \ldots$ pF/m

$$G = \tan\delta \times 2\pi \times 400 \times 10^6 \times C_{11}$$

(chap5.pdf, p.27)

1.5 Part 5 Quick Reference: Common Pitfalls

1. **Dont confuse $T_r < 2$TD with $T_r > 2$TD** for NEXT. Only in the long-line case ($T_r < 2$TD) is NEXT length-independent.

2. **Odd mode = underdriven** (lower Z); **even mode = overdriven** (higher Z). Remember: $Z_{\text{odd}} < Z_0 < Z_{\text{even}}$.

3. **FEXT = 0 in striplines** (both modes travel at the same speed). FEXT 0 in microstrips.

4. **Surface resistance** R_S is in ůs/square it does not have units of alone. $R_{\text{ac}} = R_S\sqrt{F}$.

5. **Segments formula uses TD and** T_r not just the physical length. Always check the ratio.
6. ε_e **in microstrip** uses ε_r (the bulk value), not the effective value dont iterate.
7. **AC load termination**: $R = Z_0$ AND $RC_L = 12\tau_r$ both conditions must be met.